

Mohammed Touseef Ansari

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EDUCATION

University of British Columbia

Bachelor of Science in Computer Science

Vancouver, BC

Expected May 2027

- Related Coursework: Data Structures and Algorithms, Object Oriented Programming, Computer Organization and Systems, Computer Networking, Machine Learning, Data Science Workflow, Web Scraping and Object Design.

EXPERIENCE

Markaba

Software And Automation Developer Intern

Riyadh, Saudi Arabia

July 2025 – October 2025

- Built an **end-to-end asynchronous data pipeline** using **Scrapy**, **FastAPI**, and **PostgreSQL** to ingest, sanitize, and store **32,000+ automotive listings** from **7 marketplaces** for downstream analytics and internal tooling.
- Achieved **95% crawl reliability** across protected marketplaces by implementing **adaptive throttling**, **proxy rotation**, **user-agent spoofing**, **JWT token extraction**, and **fault-tolerant retry mechanisms** to bypass anti-bot systems; orchestrated **daily workflows** across **4+ crawlers** via cron automation.
- Reduced analytics dashboard latency from **15s to under 1s** at the **p95** under concurrent load by redesigning backend query execution with **batching**, **indexing**, and **response caching**.
- Built an asynchronous **OCR microservice** using **FastAPI** and **Camelot** to extract and evaluate **SIMAH credit reports**, achieving **4× throughput improvement** and enabling configurable lending risk assessment.
- Contributed features across **frontend and backend** of the company's primary **fintech product** using **React**, **FastAPI**, and **Django** — including **role-based access control**, **audit logging**, and transaction management for lending workflows; supported **CI/CD** via **GitHub Actions** and **Docker** on **Azure**.

PROJECTS

Vectorized SQL Query Engine

Python, NumPy, Systems Design, Query Optimization

Vancouver, BC

2026

- Built a **vectorized SQL query engine** executing **SELECT**, **WHERE**, **GROUP BY**, **ORDER BY**, and **LIMIT** over **10M-row datasets** via a **tokenizer**, **Pratt parser**, logical planner, and operator-based physical execution pipeline.
- Reduced **Top-N query** latency by **12.7×** (**1.5s → 120ms**) via a **rule-based optimizer** that rewrites **Limit(Sort)** into a linear-time **TopN operator** using **NumPy argpartition**; achieved a **3800×** improvement for small-result queries with a short-circuiting **streaming Limit operator**.
- Improved wide-table scan performance by **1.5×** via **projection pushdown**; implemented **hash-based aggregation** for **GROUP BY** workloads and an **EXPLAIN visualizer** to validate optimizer rewrite rules.

Rootstock — Institutional Memory Platform

Python, LangChain, ChromaDB, OpenAI, Streamlit, D3.js

Vancouver, BC

2026

- Built a **multi-channel RAG platform** ingesting documents from **Gmail**, **Slack**, and **Google Drive** into a unified **ChromaDB vector store**, enabling cited natural language retrieval across **23+ organizational documents**.
- Implemented **dual-persona retrieval chains** using **LangChain LCEL** — a factual Q&A mode with source citations and a **mentor simulation mode** reconstructing a departed coordinator's voice from their own documents.
- Engineered an interactive **D3.js force-directed knowledge graph** with **entity co-occurrence analysis** and **single-point-of-failure detection**; added multilingual retrieval and a **spaCy NER PII redaction layer**. Placed **3rd (solo)** at **youCode @ UBC** (SAP-sponsored track); deployed on **HuggingFace Spaces**.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, R

Frameworks and Libraries: FastAPI, Django, Flask, React, Scrapy, LangChain, Camelot, NumPy, pandas, Streamlit

Databases and Systems: PostgreSQL, ChromaDB, Docker, Linux, Git, Amazon Web Services (AWS), Microsoft Azure

Concepts: REST APIs, OCR pipelines, query optimization, retrieval-augmented generation (RAG), vector search, role-based access control, observability, distributed data pipelines

Tools: GitHub Actions, NVIM, Tmux, Alacritty